NAME: NUMBER:

ELEC 361 Analog Communication Systems Midterm Exam **Fall 2023**
Prof. Dr. Oğuz Kucur **Time: 100 min.** **1.12.2023**

1)(30 p.) a) Find exponential Fourier series expansion of the signal $x(t) = \sum_{k=-\infty}^{\infty} \Pi(2000t - 2k)$. Find its power spectral density (PSD) function and sketch for $|f| \leq 3.5\ kHz$.
b) The signal $x(t)$ is applied to the input of the linear time invariant (LTI) system with the impulse response $h(t) = 20000Sinc(10000t - 100000)$. Find the transfer function of the system and sketch it. Determine if the system is distortionless or not for $x(t)$. State reasoning.
c) For the input signal $x(t) = 20Cos(4000\pi t) + 10Cos(8000\pi t)$, state if the system in part b) is distortionless or not. State reasoning. Find the PSD function, autocorrelation function and power of the output signal $y(t)$.

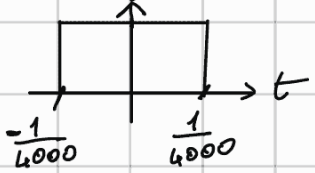
$$x(t) = \sum_{k=-\infty}^{\infty} \Pi(2000t - 2k)$$

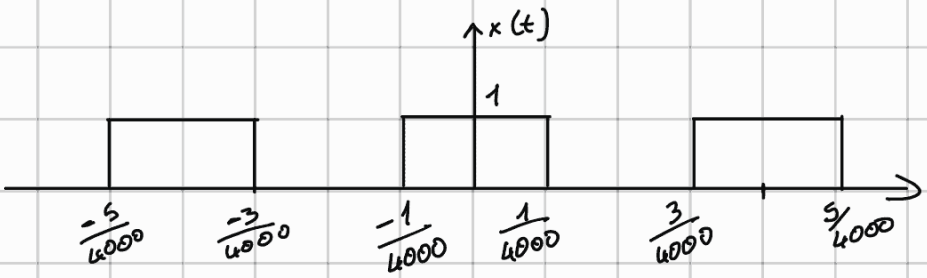
$$= \sum_{k=-\infty}^{\infty} \Pi\left(\frac{t - \frac{1}{1000}k}{1/2000}\right)$$

$$T_{period} = \frac{1}{1000} = \frac{4}{4000} = 10^{-3}\ sn$$

$$f_0 = 1khz,\ \omega_0 = 2000\pi$$

$$\Pi\left(\frac{t}{\frac{1/2000}{2}}\right) \Rightarrow$$





a) $x(t) = \sum_{n=-\infty}^{\infty} x_n e^{jn\omega_0 t}$, $x_0 = \frac{1}{T_0}\int x(t)\,dt$, $x_n = \frac{1}{T_0}\int x(t) e^{-jn\omega_0 t}\,dt$

$$x_0 = 1000 \int_{-1/4000}^{1/4000} 1\,dt = 1000 \cdot \frac{2}{4000} = \frac{1}{2}$$

$$x_n = \frac{1}{T_0}\int_{-1/4000}^{1/4000} 1 \cdot e^{-jn2\pi f_0 t}\,dt \quad (f_0 = 1000)$$

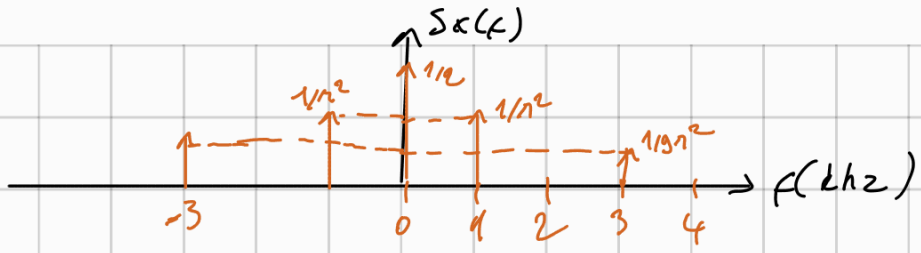
$$x_n = 10^3 \cdot \frac{-1}{jn2\pi 10^3}\left[e^{-jn2\pi \cdot 10^3 \cdot \frac{1}{4}\cdot 10^{-3}} - e^{-jn2\pi \cdot 10^3 \cdot \frac{1}{4}\cdot 10^{-3}}\right]$$

$$= \frac{1}{\pi n} \cdot \frac{1}{2j}\left[e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}\right] = \frac{1}{\pi n}\sin\left(\frac{\pi n}{2}\right), n \neq 0$$

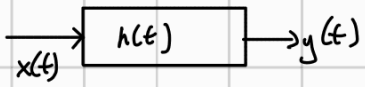
$$x(t) = \sum_{n=-\infty}^{n=\infty} x_n e^{jn2\pi f_0 t} = \frac{1}{2} + \sum_{\substack{n\neq 0 \\ n\ tek}} \frac{1}{\pi n}\sin\left(\frac{\pi n}{2}\right) e^{jn2000\pi t}$$

$$\sin\left(\frac{\pi n}{2}\right) = \begin{cases} 1, & n = 1, 5, 9, 13 \\ -1, & n = 3, 7, 11 \end{cases}$$

$$S_x(f) = \sum_n |x_n|^2 \delta(f - nf_0) = \frac{1}{2}\delta(f) + \sum_{\substack{n\neq 0 \\ n\ tek}} \frac{1}{\pi^2 n^2}\delta(f - n1000)$$



b)



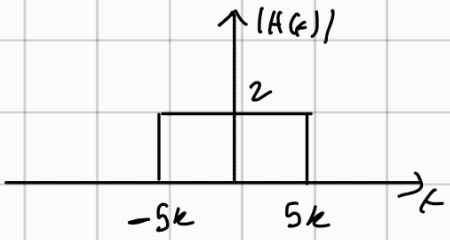
$h(t) = 20000 \operatorname{sinc}(10k(t-10))$

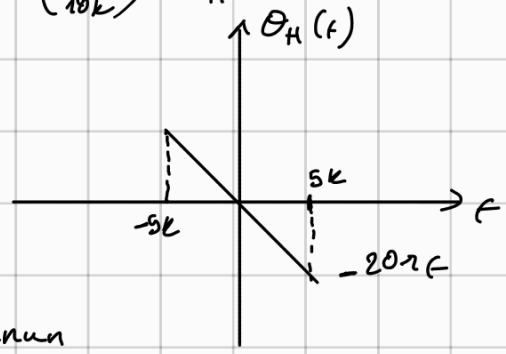
$Az \quad z = 10k$

$A = 2$

$\Leftrightarrow H(f) = 2\Pi\left(\frac{f}{10k}\right) e^{-j2\pi f 10}$

$|H(f)| = 2\Pi\left(\frac{f}{10k}\right) \quad \Theta_H(f) = -2\pi 10 f$



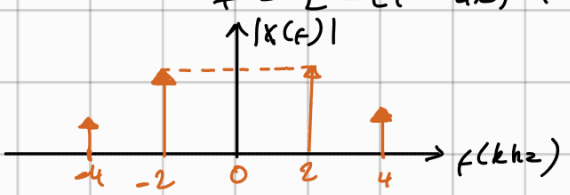


$\Rightarrow |H(f)|$, $x(t)$'nin spektrumunun tamamında $|k| \neq 0$ olmadığından bu işaret bozulmasız değildir

c) $x(t) = 20\cos(2\pi 2kt) + 10\cos(2\pi 4kt)$

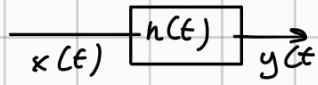
$X(f) = 10[\delta(f-2k) + \delta(f+2k)]$

$+ 5[\delta(f-4k) + \delta(f+4k)]$



$|H(f)|$, $X(f)$'in BG'si içinde sabit, $\Theta_H(f)$'de değiştiği için sistem bu işaret için bozulmasızdır.

$k = 2, \; t_d = 10$



$y(t) = 2x(t-10)$

$S_y(f) = |H(f)|^2 \cdot S_x(f)$

$S_x(f) = 100\delta(f-2000) + 100\delta(f+2000) + 25\delta(f-4k) + 25\delta(f+4k)$

$S_y(f) = 400\delta(f-2k) + 400\delta(f+2k) + 100\delta(f-4k) + 100\delta(f+4k)$

$\downarrow F^{-1}$

$R_y(z) = 800\cos(2\pi 2kz) + 200\cos(2\pi 4kz)$

1. yol $P_y = R_y(0) = 800 + 200$

2. yol $P_y = \int_{-\infty}^{\infty} S_y(f)\,df = 2 \cdot 400 + 2 \cdot 100 = 1000\,W$